

ASSIGNMENT2

1. What are the two main factors, which decide the efficiency of the algorithm?
2. Differentiate priori and posteriori analysis of the algorithm
3. What do you understand about time and space complexity?
4. With the use of the examples clearly explain commonly used asymptotic notations to calculate the running time complexity of an algorithm.
5. Calculate the time and space complexity of the following code segments.

i. Algorithm swap(a,b)

start

temp :=a

a:=b

b:=temp

End

ii. below is the function which return the sum of the elements in the array:

```
int sum(a,n){  
    int i , sum=0;  
    int a[n]={1,2,3,4,5};  
    for(i=0; i<n; i++){  
        sum =sum+a[i];  
    }  
    return sum;  
}
```

iii. following code segment will calculate the sum of 2 matrices:

```
int sum(a,b,n){
```

```
int i, j, sum[n];
for (i=0; i<n; i++){
    for (j=0; j<n; j++){
        sum[i][j] = a[i][j];
    }
}

return sum;
}
```

- iv. calculate time and space complexity of the following for loop:

```
int count = 0;
for(i=1; i<n; i=i*2){
    count++;
}
```

6.